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java.lang.invoke

Class ConstantCallSite

java.lang.Object
  java.lang.invoke.CallSite
    java.lang.invoke.ConstantCallSite

```
public class ConstantCallSite
extends CallSite
```

A `ConstantCallSite` is a `CallSite` whose target is permanent, and can never be changed. An `invokedynamic` instruction linked to a `ConstantCallSite` is permanently bound to the call site's target.

Constructor Summary

Constructors

Modifier	Constructor	Description
	ConstantCallSite(**MethodHandle** target)	Creates a call site with a permanent target.
protected	**ConstantCallSite**(**MethodType** targetType, **MethodHandle** createTargetHook)	Creates a call site with a permanent target, possibly bound to

		the call site itself.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`MethodHandle`	`dynamicInvoker()`	Returns this call site's permanent target.
`MethodHandle`	`getTarget()`	Returns the target method of the call site, which behaves like a `final` field of the `ConstantCallSite`.
`void`	`setTarget(MethodHandle ignore)`	Always throws an `UnsupportedOperationException`.

Methods inherited from class java.lang.invoke.CallSite

`type`

Methods inherited from class java.lang.Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Constructor Detail

ConstantCallSite

```
public ConstantCallSite(MethodHandle target)
```

Creates a call site with a permanent target.

Parameters:

`target` - the target to be permanently associated with this call site

Throws:

`NullPointerException` - if the proposed target is null

ConstantCallSite

```
protected ConstantCallSite(MethodType targetType,
                           MethodHandle createTargetHook)
                    throws Throwable
```

Creates a call site with a permanent target, possibly bound to the call site itself.

During construction of the call site, the `createTargetHook` is invoked to produce the actual target, as if by a call of the form `(MethodHandle) createTargetHook.invoke(this)`.

Note that user code cannot perform such an action directly in a subclass constructor, since the target must be fixed before the `ConstantCallSite` constructor returns.

The hook is said to bind the call site to a target method handle, and a typical action would be `someTarget.bindTo(this)`. However, the hook is free to take any action whatever, including ignoring the call site and returning a constant target.

The result returned by the hook must be a method handle of exactly the same type as the call site.

While the hook is being called, the new `ConstantCallSite` object is in a partially constructed state. In this state, a call to `getTarget`, or any other attempt to use the target, will result in an `IllegalStateException`. It is legal at all times to obtain the call site's type using the `type` method.

Parameters:

`targetType` - the type of the method handle to be permanently associated with this call site

`createTargetHook` - a method handle to invoke (on the call site) to produce the call site's target

Throws:

`WrongMethodTypeException` - if the hook cannot be invoked on the required arguments, or if the target returned by the hook is not of the given `targetType`

`NullPointerException` - if the hook returns a null value

`ClassCastException` - if the hook returns something other than a `MethodHandle`

`Throwable` - anything else thrown by the hook function

Method Detail

getTarget

```
public final MethodHandle getTarget()
```

Returns the target method of the call site, which behaves like a `final` field of the `ConstantCallSite`. That is, the target is always the original value passed to the constructor call which created this instance.

Specified by:

`getTarget` in class `CallSite`

Returns:

the immutable linkage state of this call site, a constant method handle

Throws:

`IllegalStateException` - if the ConstantCallSite constructor has not completed

See Also:

`ConstantCallSite`, `VolatileCallSite`, `CallSite.setTarget(java.lang.invoke.MethodHandle)`, `getTarget()`, `MutableCallSite.getTarget()`, `VolatileCallSite.getTarget()`

setTarget

```
public final void setTarget(MethodHandle ignore)
```

Always throws an `UnsupportedOperationException`. This kind of call site cannot change its target.

Specified by:

`setTarget` in class `CallSite`

Parameters:

ignore - a new target proposed for the call site, which is ignored

Throws:

`UnsupportedOperationException` - because this kind of call site cannot change its target

See Also:

`CallSite.getTarget()`, `setTarget(java.lang.invoke.MethodHandle)`, `MutableCallSite.setTarget(java.lang.invoke.MethodHandle)`, `VolatileCallSite.setTarget(java.lang.invoke.MethodHandle)`

dynamicInvoker

```
public final MethodHandle dynamicInvoker()
```

Returns this call site's permanent target. Since that target will never change, this is a correct implementation of `CallSite.dynamicInvoker`.

Specified by:

`dynamicInvoker` in class `CallSite`

Returns:

the immutable linkage state of this call site, a constant method handle

Throws:

`IllegalStateException` - if the ConstantCallSite constructor has not completed

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